Predicting Singapore HDB Rental Prices under Temporal Distribution Shift

Ox Alpha (Group 6)
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Abstract—
Writing prompt: Write approximately 150–180 words after the final experiments are complete. State the practical rental-prediction problem, the dataset and temporal challenge, the analysis and preprocessing approach, the evaluated methods, and the selected final model. Give its verified evaluation results and the most important interpretation. Describe completed work in the past tense. Do not present progress-report scores as final results.

Index Terms—HDB rental prediction, temporal validation, feature engineering, regression, spatial auxiliary data

I. Introduction

A. Motivation and Problem Context

Writing prompt: Explain how estimating HDB rents supports rental-budget planning and pricing decisions. Identify the intended users and why prediction is challenging when market conditions change. Use verified sources for external market or policy claims. Kaggle ranking should not be the primary motivation.

B. Research Questions and Contributions

Writing prompt: Organise the paper around the following questions. State contributions that the completed experiments actually support, rather than claiming that a new algorithm was invented.

1) How do temporal, spatial and property patterns inform preprocessing and prediction of rental prices?
2) What predictive value do auxiliary features add beyond the core rental attributes under a controlled comparison?
3) Which completed model improvements produce the most reliable performance across later time periods, and where does the final model fail?

II. Data and Problem Formulation

A. Prediction Task and Core Data

Writing prompt: Define supervised regression from rental attributes to MONTHLY_RENT in SGD. Summarise the supplied inputs, labelled data and test data. Distinguish raw sample counts from the counts used by the final training pipeline. The goal is prediction, not estimation of causal effects.
Writing prompt: Refer to Table I. Explain that a small part of the test data overlaps the final training month, while most records are later. Do not describe every test record as strictly later than all training observations.

TABLE I
Overview of the supplied rental data.

Property	Training data	Test data
Raw records	150,000	50,000
Rental months	Jan. 2021–Mar. 2025	Mar. 2025–Jul. 2026
Supplied predictors	10	10
Target labels	Available	Unavailable
Final modelling records	TBD	50,000

B. Auxiliary Data Sources

Writing prompt: Introduce HDB block metadata, MRT stations, schools, shopping malls, COE prices and stock prices. Identify which sources were evaluated and which were retained in the final model. Investigation or feature construction alone does not establish predictive value. Cite the actual dataset sources.

III. Exploratory Data Analysis and Preprocessing

A. Target Distribution and Data Quality

Writing prompt: Describe the target distribution and raw completeness. Explain why constant FURNISHED and FEE fields were removed. Discuss the IQR-flagged extremes and the final outlier policy. Do not equate an unusual rent with a proven recording error. State whether the final model used raw rent or a transformed target and support the decision with completed tests.

B. Temporal Patterns and Distribution Shift

Writing prompt: Explain the rapid rental growth during 2021–2023 followed by much smaller changes in 2024 and early 2025. Connect this change to the validation design and the final representation of time. Label early-2025 statistics as partial-year statistics. Figure 1 should establish this finding.

C. Spatial and Property Patterns

Writing prompt: Summarise rent differences by town, flat type and floor area. Explain the confounding between time, location and property size. Show only the EDA evidence that supports a modelling decision. Rent per square metre may be used descriptively but must not enter predictors because it contains the target. Put additional town and flat-model plots in the appendix.

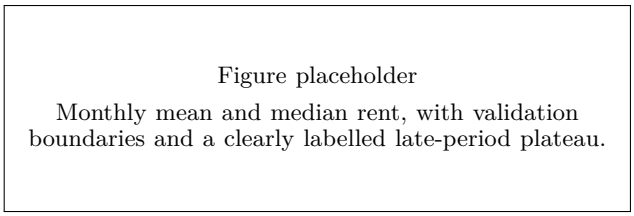

Fig. 1. Rental-price patterns over the training period. Replace this placeholder with a plot and specify the sample population in the caption.

TABLE II
EDA findings and the corresponding preprocessing decisions.

Finding	Decision and justification
Constant fields	Remove fields without predictive variation.
Inconsistent category labels	Normalise case, whitespace and equivalent room labels.
Temporal regime change	Describe the final trend and seasonal features, supported by evaluation.
Invalid derived lease age	Flag and treat as missing, with fold-specific imputation when required.
Repeated observations	State and justify the final duplicate policy.
Extreme targets	State the retained or filtered policy and supporting evidence.

D. Core Cleaning and Feature Construction

Writing prompt: Describe text and date normalisation, equivalent flat-type labels, location keys, lease-age handling and floor-area transformations. Explain the purpose of each retained feature group. The current core pipeline produces 19 model inputs, but update this count and the feature descriptions if the final feature set changes. Keep ROW_ID and RENT_DATE for alignment and validation, and exclude them from the predictor matrix.

E. Auxiliary Construction and Record Integrity

Writing prompt: Describe address-key joins, HDB attributes and spatial distances or counts. Explain how historical availability was handled for MRT and other snapshot data. For economic variables, describe aggregation, calendar lags and information availability. Report verified coverage and missingness for sources actually used.

Writing prompt: State the final exact-duplicate and repeated-feature policies. The current core removes 641 exact repeats and retains 149,359 rows. An earlier auxiliary file collapsed feature vectors and averaged rents. The corrected Ridge comparison uses the same individual core rows and original targets. Report the final policy, preserve comparable evaluation populations, and explain any weighting if records were aggregated. Do not compare scores computed against original targets with scores computed against group means.

IV. Data Mining Methods and Model Development

A. Reference Predictor and Ridge Baseline

Writing prompt: Describe the historical town–flat-type median reference and its fallback rules. Explain why Ridge was selected as a fast baseline for sparse, correlated predictors. State its feature set and target representation. Describe how it was applied rather than reproducing textbook derivations.

B. Candidate Models

Writing prompt: Describe the alternative models actually evaluated after the progress report. Possible candidates include boosting models, but do not list a method as an experiment unless it was run. For each model, explain its suitability, categorical handling, target representation and principal configuration. Add verified references for advanced methods. Remove unused candidates.

C. Completed Model Improvements

Writing prompt: Use this subsection for the improvements completed before submission. For each change, connect an observed weakness to a specific modification and its controlled evaluation. Report unsuccessful changes when they explain why a component was excluded. Do not write this subsection as a plan.

1) Temporal Representation and Calibration: Writing prompt: Describe any completed changes to linear time trends, piecewise or capped trends, training windows, recency weights or calibration. Explain how cutoffs and other choices were selected using historical validation. State which approach the final model retained. Delete this heading if no such experiment was completed.

2) Feature Interactions and Auxiliary Information: Writing prompt: Describe completed tests of time–location or property interactions, alternative categorical representations, and additional auxiliary feature groups. Identify retained and rejected features. Distinguish a plausible explanation from an effect demonstrated by ablation.

3) Tuning and Model Combination: Writing prompt: Describe completed tuning, early stopping or ensemble experiments. If an ensemble was used, specify its component models, training protocol and how its weights were selected without using evaluation labels. Remove this heading when the final method is a single model with no relevant experiment.

D. Final Model Specification

Writing prompt: Give the selected final model or ensemble, exact feature groups, target transformation, hyperparameters and training procedure. Explain the selection criterion. This subsection must describe the implemented final system, not merely the highest-scoring intermediate run. State the final refitting procedure and any post-processing used for test predictions.

TABLE III
Existing chronological validation windows. Update to match the final evaluation protocol.

Fold	Training through	Validation period
1	Jun. 2023	Jul.–Dec. 2023
2	Dec. 2023	Jan.–Jun. 2024
3	Jun. 2024	Jul. 2024–Mar. 2025

V. Experimental Setup

A. Chronological Validation

Writing prompt: Explain why evaluation must train on earlier rentals and predict later rentals. Table III shows the existing expanding windows. Replace or supplement them if the final protocol changes. State the prediction horizon and any gap. Explain how final performance was evaluated separately from tuning, or explicitly disclose reuse of the tuning folds.
Writing prompt: The current validation blocks span 6, 6 and 9 months, while the maximum test prediction distance is 16 months after the last training month. Describe any completed longer-horizon evaluation, or acknowledge the horizon limitation. A fold already used to choose models is not an untouched holdout.

B. Training-Fold Preprocessing and Leakage Controls

Writing prompt: Describe how numerical imputation, scaling, categorical vocabularies and infrequent-category grouping were fitted inside each historical training fold. Explain any model-specific exceptions. Exclude target-derived inputs. State how unknown categories and auxiliary missingness were handled.

C. Hyperparameter Selection and Reproducibility

Writing prompt: Report the actual search strategy, search space, validation criterion, random seeds, software versions and relevant resource constraints. The existing Ridge search uses alpha in {0.1, 1, 10, 100}; update this if expanded. Summarise decisive settings in the main text and put the full grid in the appendix. Every result must be traceable to a saved experiment configuration.

D. Metrics and Controlled Comparisons

Writing prompt: Use RMSE as the primary metric and MAE and R-squared as supporting diagnostics. Define whether results are unweighted means of fold metrics or pooled validation metrics. Use identical record IDs and original labels for feature comparisons. State the tuning protocol for model comparisons. If uncertainty is reported, specify how temporal dependence was handled.

VI. Final Results and Discussion

A. Overall Model Comparison

Writing prompt: Fill Table IV with the final verified results from one consistent evaluation protocol. Compare the reference, baseline, improved model and selected final configuration. Explain why the final model was selected, including performance consistency and relevant trade-offs. All values below remain TBD until the final runs are frozen.

Figure placeholder
Per-fold RMSE for the baseline and final model.

Fig. 2. Performance across chronological validation periods. Replace the placeholder with results from the final experiment runs.

Figure placeholder
Monthly actual and predicted mean rent for the baseline and final model on the same latest evaluation period.

Fig. 3. Temporal calibration of the baseline and final model. Report mean bias and the validation period in the completed caption.

B. Performance across Time Periods

Writing prompt: Report per-fold results and explain differences between market periods. A lower average alone is insufficient to establish temporal robustness. Figure 2 should compare the principal methods across the same validation windows.

C. Ablation Studies

Writing prompt: Quantify the contribution of completed improvements by adding or removing one component while keeping the evaluation population fixed. Select the most informative feature, temporal and modelling ablations. Table V is a template, not a required list of experiments.
Writing prompt: The progress report's approximately 0.1% HDB/MRT gain is an intermediate Ridge finding. Re-evaluate its contribution under the final model and describe tiny differences proportionately. Do not call a difference statistically significant without an appropriate uncertainty analysis.

D. Error Analysis and Model Interpretation

Writing prompt: Analyse the final model's errors by month, flat type, town and rent range. Report subgroup sample sizes, RMSE or MAE, and signed bias. Define residual as actual minus predicted. Show whether the final model still overpredicts the late-period plateau and compare with the Ridge baseline. Explain poor cases with observed evidence, while distinguishing unobserved causes and hypotheses. Treat correlated coefficients or feature attributions as predictive descriptions, not causal effects.

E. Final Test Predictions and Submission

Writing prompt: Identify the final refit, submission configuration and 50,000 predictions aligned to the sample-submission IDs. Report an actual competition score only if available, specifying public/private split and the selected submission. Do

TABLE IV
Final comparison under the common evaluation protocol. Replace generic model names and remove rows for experiments not completed.

Method	Feature set	RMSE (SGD)	MAE (SGD)	R^2	RMSE reduction (%)
Historical median	Town and flat type	TBD	TBD	TBD	—
Ridge baseline	Core	TBD	TBD	TBD	TBD
Ridge baseline	Core + HDB/MRT	TBD	TBD	TBD	TBD
Improved model	Specify evaluated inputs	TBD	TBD	TBD	TBD
Final selected model	Specify final inputs	TBD	TBD	TBD	TBD

Specify the score aggregation and evaluation population. Define percentage reduction relative to the Core Ridge baseline. If the improved and final models are identical, combine those rows. Use a clearly separate column for any external competition score.

TABLE V
Controlled ablation of the final model.

Configuration	RMSE (SGD)	ΔRMSE
Full final configuration	TBD	0.00
Without auxiliary spatial features	TBD	TBD
Without the selected temporal change	TBD	TBD
Without a tested interaction/component	TBD	TBD

Define ΔRMSE as the ablated configuration's score minus the full configuration's score. Remove rows for components not tested.

not infer test accuracy from prediction ranges or compare a competition score directly with a different internal validation population.

F. Limitations of the Final Findings

Writing prompt: Discuss limitations supported by the completed analysis: missing unit-level attributes, small subgroups, duplicate-record uncertainty, historical snapshot accuracy, correlated features and evaluation-horizon coverage. Explain how these constrain the claims about the selected model. Keep this as an assessment of the final evidence, without a future-outlook or planned-experiments section.

VII. Conclusion

Writing prompt: Answer the research questions using the completed final results. Name the selected model, quantify its performance relative to the baseline, state which improvements and auxiliary features proved useful, and summarise the most important error or limitation. End with the practical interpretation of the findings. Do not end with a list of work the team intends to do.

References

[1] Placeholder: verified citation for the course rental dataset and the auxiliary data sources actually used.

[2] Placeholder: verified reference for Ridge regression, if cited.

[3] Placeholder: original paper or authoritative reference for the advanced method actually used in the final model.

Appendix A
Team Workload Breakdown

Writing prompt: Record actual completed contributions. The names, IDs and team name come from the supplied progress slides. Confirm them before submission. Only Sun Zekun's core-preprocessing responsibility is prefilled from the conversation; complete all other entries from the team's actual workload.

Appendix B
Supplementary Details

Writing prompt: Use sparingly. Include only the full hyperparameter search/configuration table, a compact feature dictionary, or supplementary EDA/error tables that help reproduce or elaborate the main results. Do not move the final model definition, validation design, main results or essential interpretation out of the main text. The paper must remain comprehensible without this appendix.

TABLE VI
Team workload and report contributions.

Member	Student ID	Main project responsibilities	Report contributions
Li Chengxuan	A0352428A	TBD	TBD
Mao Huanqing	A0257997Y	TBD	TBD
Sun Zekun	A0351314R	Core data preprocessing; complete other actual responsibilities	Core preprocessing; complete actual writing/review tasks
Zhang Ziqing	A0223145R	TBD	TBD